

We discuss dielectric haloscopes as a way to detect axions, which are a promising candidate for dark matter. Building a dielectric haloscope is a medium term goal of the Broadband Reflector Experiment for Axion Detection (BREAD) collaboration at Fermilab. We discuss the theory behind this new design and potential advantages of the new design. In order to build an accurate dielectric haloscope, we must first build an apparatus to accurately measure the complex relative permittivity and loss tangent of dielectric materials. We then present a summary of the major relative permittivity measurement methods and explain why we chose a waveguide based system. We give an overview of the major ways to extract the relative permittivity from measured scattering parameters (obtained from a Vector Network Analyzer) including the method of least squares, the Nicholson-Ross-Weir method, and the NIST iterative method. Our waveguide based relative permittivity measurement setup is presented. We discuss Network Analyzer calibration and simulations of the measurement setup that were run in Ansys HFSS. Simulated and measured S-parameter data from a WR-42 waveguide are shown. Finally, we present a measurement of the permittivity of Rogers TMM3 materials as well as simulations of the measurement process for Rogers TMM4 and Teflon materials.